

MMS210 Connected fleets in data-driven engineering – Lab project report

Group 2 (Truck)

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Introduction

This project looks at the vibrations that truck drivers feel on real roads. It does this by creating computer versions of different road surfaces and then seeing how those surfaces affect the truck driver's seat suspension. Because a driver's comfort depends on how well the seats absorb bumps, knowing how the vibrations behave is a key step toward giving a smoother ride and reducing the tiredness that builds on long journeys.

We pay special attention to vibrations between 4 Hz and 8 Hz, because ISO 2631—the global rulebook for whole-body shaking—shows this band affects the human body the most. When a driver sits in these slow shakes for long hours, comfort drops and alertness reduces, and health problems may build up over time. Tuning the seat suspension to trim these 4-8 Hz waves is essential for driver safety and comfort, and for efficient truck operation.

To explore the problem, the work builds a three-degree-of-freedom mathematical model, that represents how the truck, the seat, and driver move together. Real-world numbers were gathered with a data logger fixed to the driver's seat. Over a 30 km test run that mixed smooth, average, and rough road stretches, the logger stored acceleration, yaw, and GPS data. Those

measured values were then set side-by-side with the model's predictions to judge how trustworthy the model is.

By using computer simulation, frequency study, and tests against ISO rules, this research reveals how each type of road surface changes what the driver feels and shows ways to fine-tune the seat suspension. The aim is to build a repeatable method that truck makers—Volvo, Scania, MAN, Daimler, and others—can follow to craft trucks that stay comfortable and dependable in any driving environment.

In the end, the work helps original equipment manufacturers create smart, adaptive seat suspensions and gives them a way to test, prove, and spread vibration-control ideas across their line-ups, keeping ride quality steady from one production model to the next.

Methodology



Fig 1. Group photo with the Volvo truck used for the project

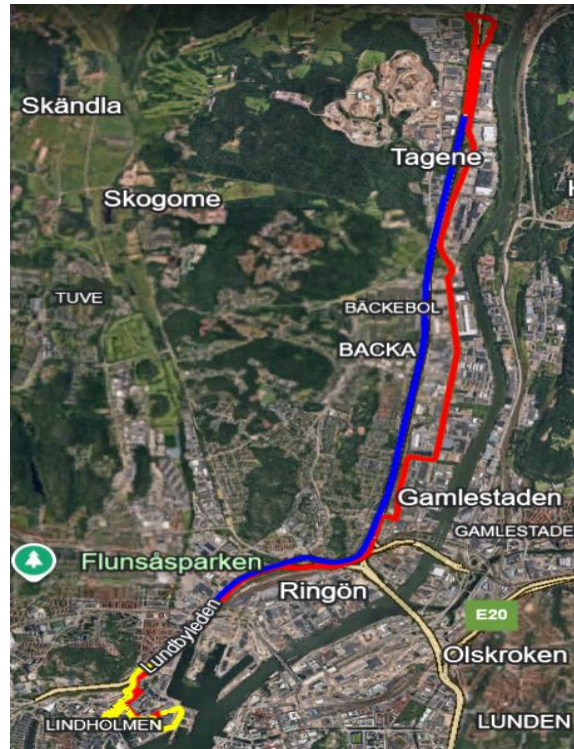


Fig 2. This is the trajectory map followed for the collection of data. Total distance covered for the collection of data is around 30 kms in Gothenburg area.

Vehicle Overview

- The vehicle used is a Volvo FH16, 780 HP truck.

Suspension System

The vehicle uses air suspension for both

- Seat damping
- Vehicle suspension

Benefits of the air suspension

- Improved ride comfort
- Enhanced vehicle stability
- Automatically adjusts under varying load and road conditions.

Mathematical Model Assumption

- 3-DoF seat-cab model built in MATLAB/Simulink.
- Quater- car models have been added for cross checking.
- Sprung mass (m_s) - Represents the body/chassis mass.

- Unsprung mass (m_{us}) - Represents the wheel and axel mass.
- Road profiles are assumed to be smooth to rough surfaces.

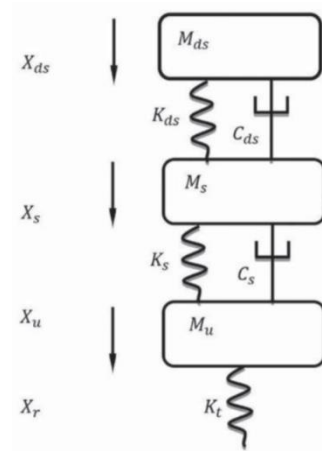


Fig 3. Model for 3 – DOF.

Process chart

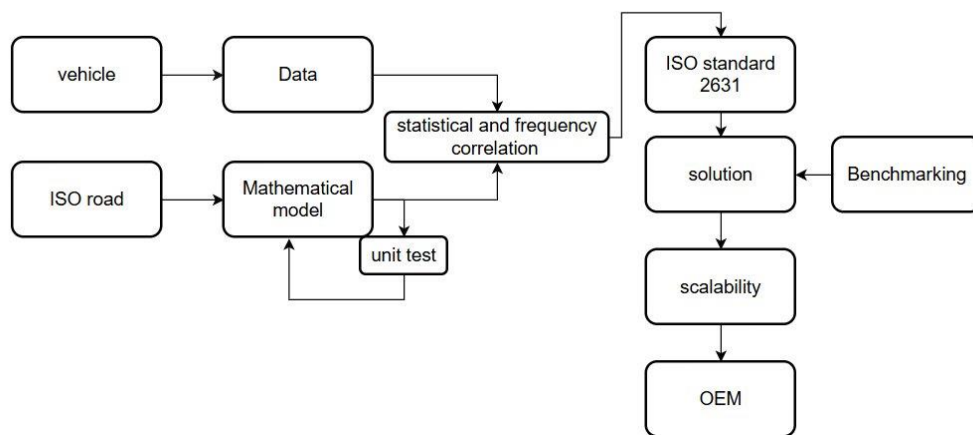


Fig 4. Process chart of vehicle comfort assessment and modeling process

This flow chart highlights the vehicle comfort assessment and modeling process using data analysis and standards such as ISO 2631.

1. Data collection

- We logged raw data from the truck — **seat acceleration, yaw angle, GPS location** and other signals. We keep the yaw angle (how much the truck turns left or right) so we can line up the accelerometer axes during a turn, feel the same twist the driver feels, and tag cornering moves to keep our comfort numbers accurate.



Fig 4a. Logger picture on Volvo FH at driver's armrest (*We didn't have a better picture*)

2. Mathematical Model & Testing

- ISO road shapes are used to copy how the truck reacts, giving a fair comparison.
- A digital model is created to reflect how the truck moves in fixed, known conditions.
- We have tested the model on its own to make sure its results are right.
- We match the measured data against the model, using stats and frequency-domain links to prove and polish the model further.

3. Standards & Solution

- ISO standard 2631 defines how to evaluate human exposure to whole body vibration.

- A solution or method for improving ride comfort is developed based on the model and standard.
4. Benchmarking, Scalability & Deployment
- The developed solution is compared with baseline models and competitor vehicles.
 - The solution is evaluated for scalability can it be applied to different types of vehicles and conditions?
 - The final validated and scalable solution is delivered for the implementation for future development and manufacture of trucks.

Simulink Model

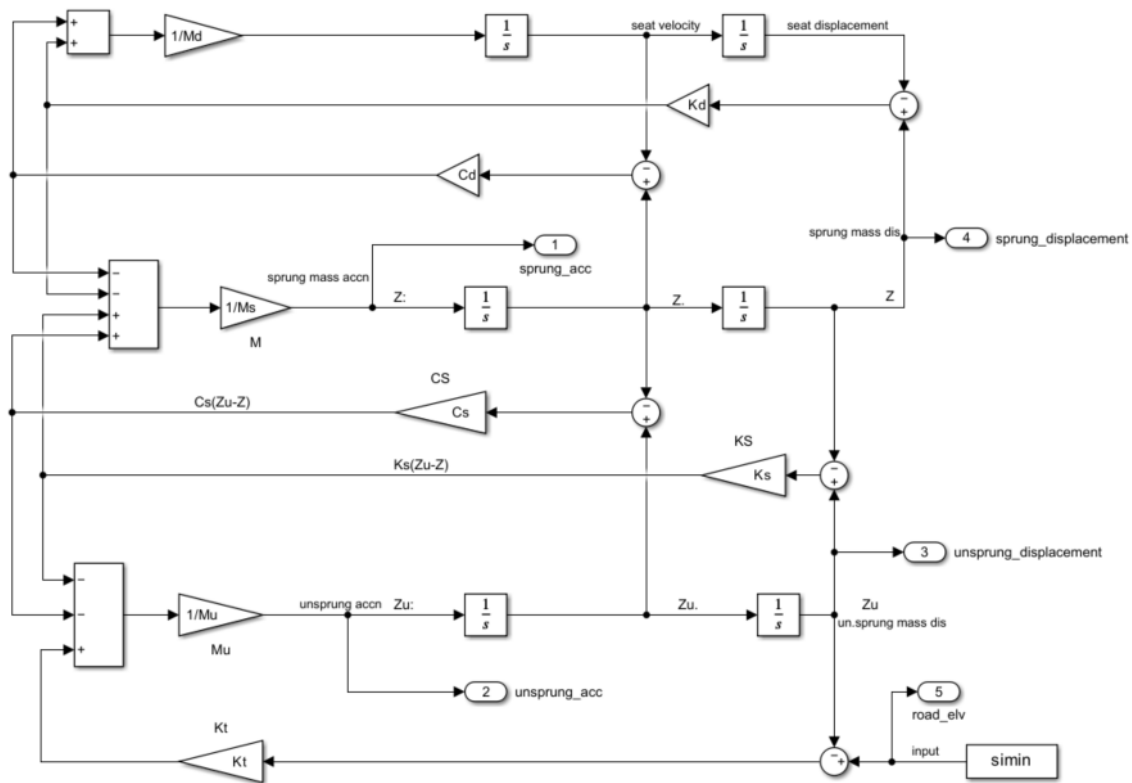


Fig 5. extended Quarter car model Simulink model.

The Simulink model of QCM is designed with all the necessary parameters of the truck taken into consideration to get suitable results. The model is designed to simulate how the truck responds to different load and road conditions generated. The road signals are the inputs, and the vibration in the driver seat is measured as output.

Results and discussion

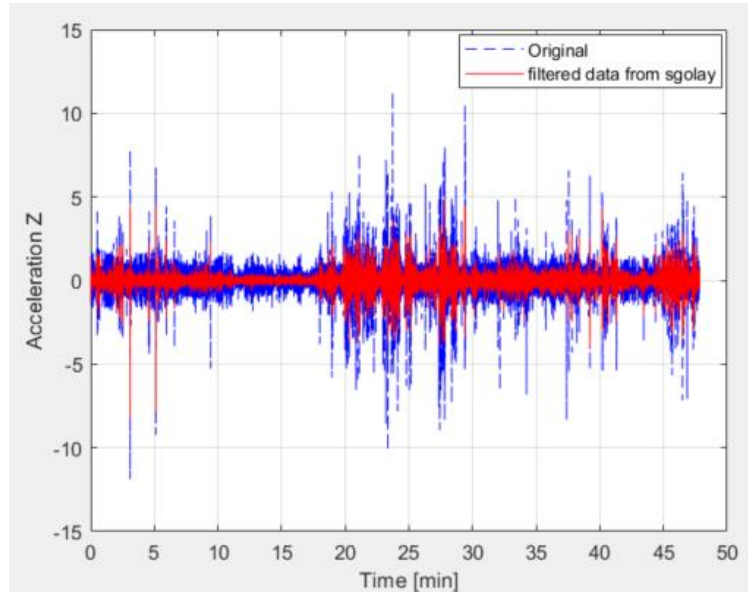


Fig 6. Raw data and Noise filtered data below

The above figure-6 shows the raw data of the entire journey for 47 minutes having different road conditions highlighted in figure 2, then entire data is filtered using sgolay to reduce the Noise without losing the actual data.

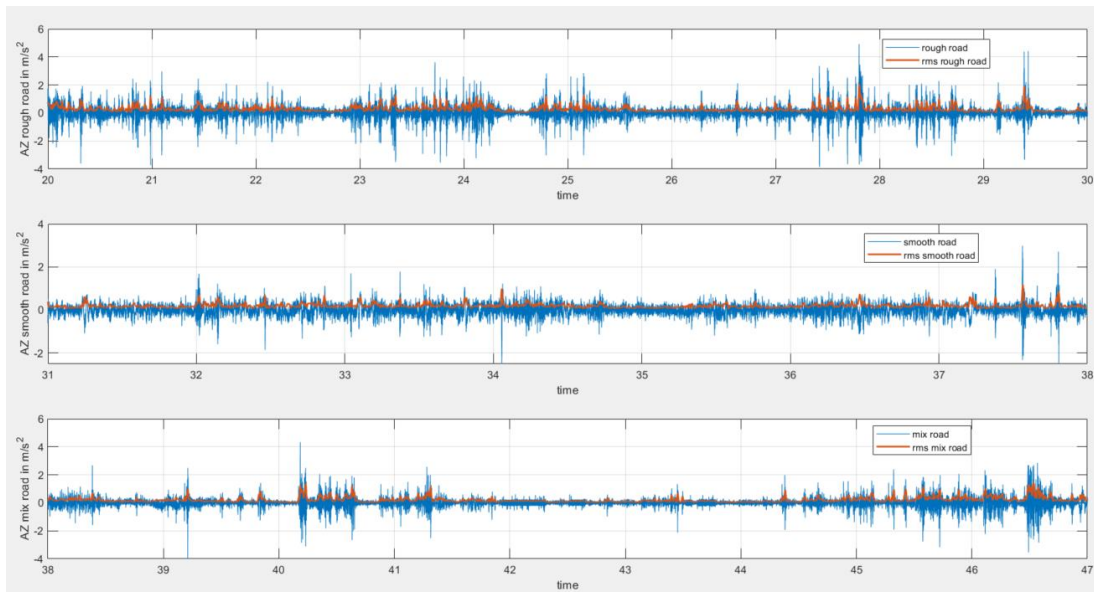
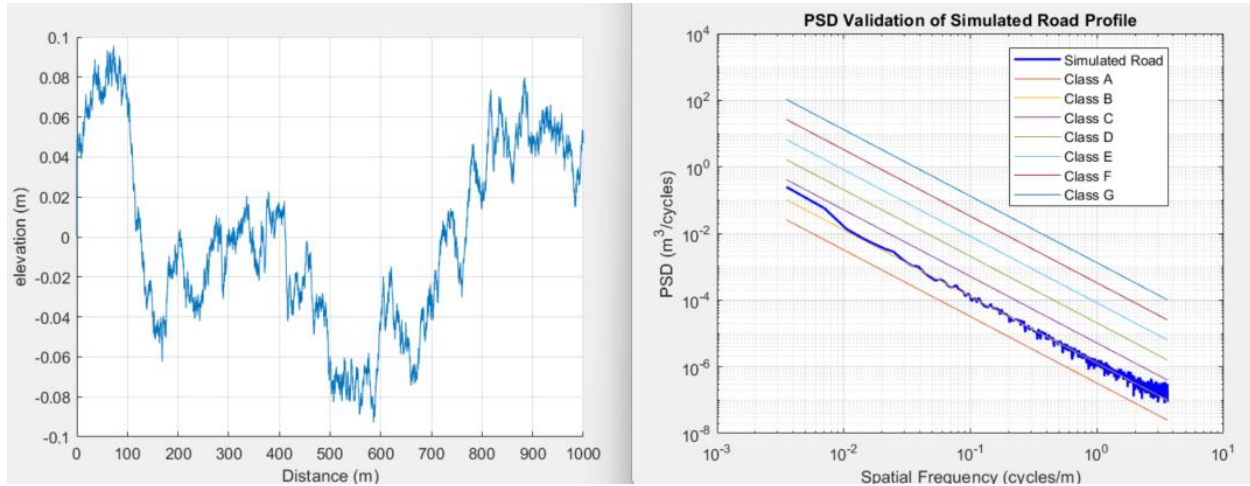


Fig 7. Road is segmented into three parts as follows and below are the rms value of data

The above figures show the segmented road raw data for different class of road and its rms plots.

Smooth Road

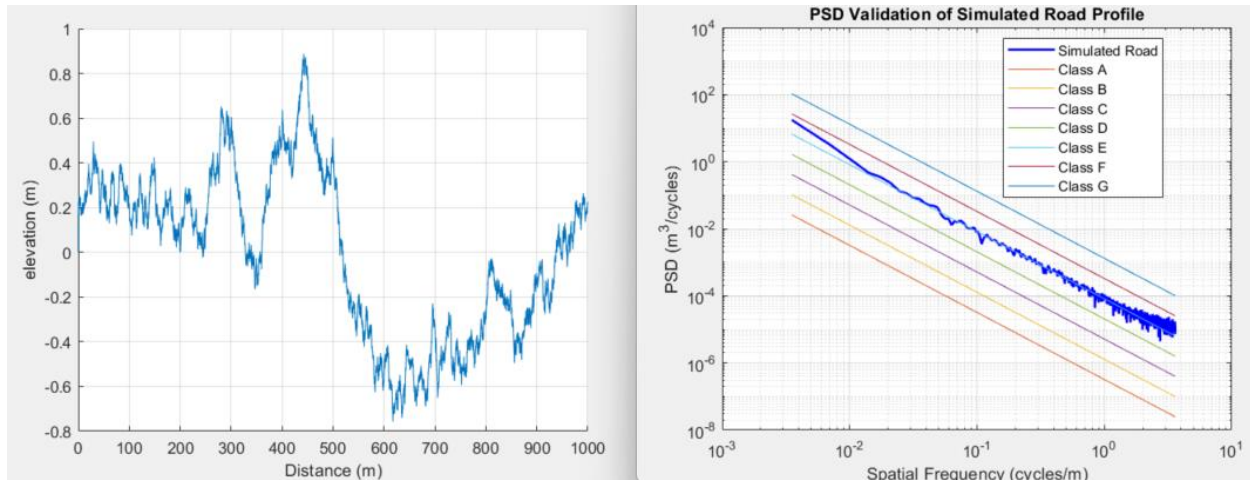
- From fig 8 road elevation(m) vs distance(m) - the elevation variations are relatively small within ± 0.1 meters. This explains the well-maintained road surface typical of class B according to ISO 8608.
- PSD(m^3/cycles) vs spatial frequency(cycles/m) - the blue curve represents simulated road data. It aligns closely with Class B. PSD reduces steeply with the spatial frequency, indicating low-frequency, small-amplitude road irregularities.



Smooth road. Fig 8 elevation(m) vs distance(m) and PSD(m^3/cycles) vs spatial frequency (cycles/m)

Rough Road

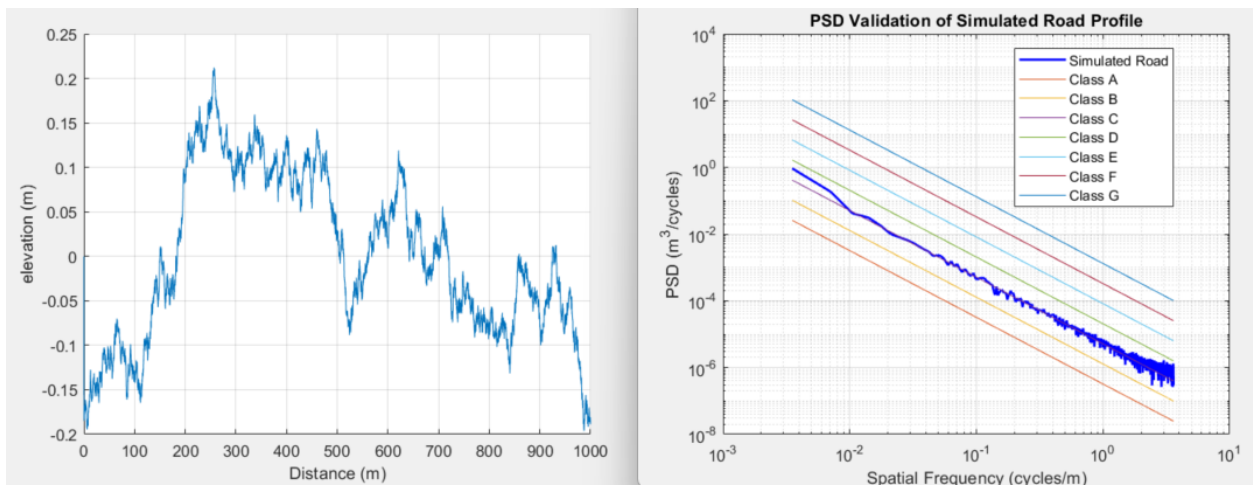
- From fig 9 road elevation(m) vs distance(m) - the elevation varies sharply between -0.8 and +0.8 meters. This explains the off-road conditions.
- PSD(m^3/cycles) vs spatial frequency(cycles/m) - the blue curve represents simulated road data. It overlaps closely with Class E. PSD reduces steeply with the wide range of frequencies, which stress the suspension system aggressively.



Rough road. Fig 9 elevation(m) vs distance(m) and PSD(m^3/cycles) vs spatial frequency (cycles/m)

Mix Road

- From fig 10 road elevation(m) vs distance(m) - the elevation varies sharply between -0.18 and +0.22 meters. The middle section distance from 200 to 600 m shows higher elevation and steeper gradient which explains rough road. The start and end sections are relatively flat which explains a smoother path.
- PSD(m^3/cycles) vs spatial frequency(cycles/m) - the blue curve overlaps closely between class B, C & D explains the mixed nature of the road.



Mix road. Fig 10 elevation(m) vs distance(m) and PSD(m^3/cycles) vs spatial freq(cycles/m)

FFT overall considering class D.

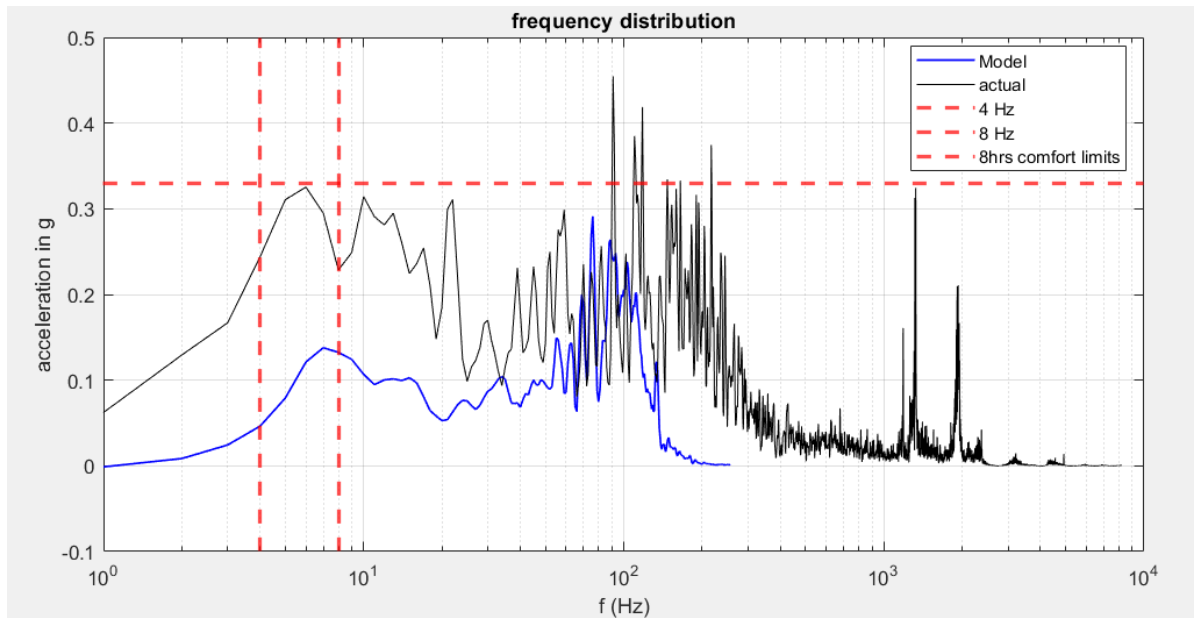


Fig 11 acceleration(g) vs frequency distribution (Hz)

From Fig. 11, we see that frequencies between 4 Hz and 8 Hz are important for checking ride comfort, as they mostly impact how the driver feels. The actual measured data (black line) stays under the comfort limit of 0.33 g for an 8-hour drive (horizontal red line) at these frequencies, but it goes above this limit at higher frequencies. These higher peaks show potential discomfort for drivers during long trips. On the other hand, the simulated model data (blue line) stays within the 8-hour comfort limit at frequencies between 4 Hz and 8 Hz, and at higher frequencies. This suggests that the model predicts less vibration at these critical frequencies, indicating a more comfortable ride.

The difference between simulated data and measured data can be due to several reasons. First, the simulation model might not fully consider real-world factors like actual road conditions, vehicle behavior, and how drivers handle the vehicle. All these things can strongly affect vibration levels. Also, the simulation parameters may have been idealized or simplified, causing differences between the predicted vibrations and the real measurements. Moreover, the peaks happening at different frequencies in the model compared to real measurements occur because the simulation might simplify some parts of the vehicle's suspension system or its interaction with the road. These simplifications can shift the frequency response. This shows clearly that we need to improve the simulation model, so it better represents real driving conditions. Doing this can help the simulation match real-world data more closely.

Considering a day (8 hours) drive for comfort level, the values below the model perform well in

smooth and mixed road types, but there is improvement needed in rough road. This might be because of model limitations and assumptions made.

Road type	Rms value of segmented data	Avg Speed in m/s	Rms value model	Error in estimation
Smooth (highway)	0.266	21.5	0.226	4% under estimation
Rough road	0.414	16.1	0.772	35% over estimation
mix	0.346	13.2	0.377	3% under estimation

Benchmarked data from already existing solutions:

Manufacturer and Model	Volvo FH16 780HP	Scania R 730	Kenworth T680
Vehicle Mass (LADEN)	26000 kg	25760 kg	27000 kg
Mass Distribution (Front/Rear)	29/71	35/65	45/55
Front Suspension (Type)	3 leaf parabolic spring	3x29 parabolic front springs	Taperleaf with shocks
Rear Suspension (Type)	4-bag air suspension on tandem rear axles	Scania 4 bag air suspension	four-bag air-ride suspension
Front suspension Spring Stiffness	540 kN/m	450 kN/m	358 kN/m
Rear suspension Spring Stiffness	2300 kN/m	2000 kN/m	1800 kN/m
Seat Mass	35 kg	30 kg	40 kg
Tire Dimension Front	315/80 R22.5	295/80 R22.5	315/70 R22
Tire Dimension Rear	315/80 R22.5	295/80 R22.5	315/70 R22
Tyre vertical stiffness	1300 kN/m	1273 kN/m	1350 kN/m

To evaluate driver comfort in heavy trucks, we benchmarked three truck models Volvo FH16 780HP, Scania R 730, and Kenworth T680 focusing on key parameters such as suspension stiffness, mass distribution, seat mass, and tire characteristics. Based on these and taking into consideration ISO 2631 standards for whole-body vibration (WBV), we provide the following analysis and improvement strategies.

The truck suspension systems were modelled as mass-spring systems to estimate natural frequencies:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Where k is spring stiffness and m is effective axle mass. Calculated sprung mass frequencies are:

Truck Model	Front Suspension Frequency (Hz)	Rear Suspension Frequency (Hz)
Volvo FH16 750	2.39	5.63
Scania R 730	2.25	5.54
Kenworth T680	1.71	5.56

According to ISO 2631, the human body is most sensitive to vertical vibrations in the 4–8 Hz range. As we can see, rear suspension frequencies of all models fall within this sensitive band, which may contribute to increased discomfort for the driver.

Seat masses vary between 30–40 kg. While higher mass can aid in absorbing energy, seat comfort depends heavily on damping and isolation. To reduce vibration transmission to the driver we can use the following:

- Use vibration-isolating seat mounts.
- Employ active seat suspension systems.
- Apply PID-based control algorithms to counteract vibrations in real-time using data from the data logger.

All models feature tire vertical stiffness between 1270–1350 kN/m. Stiffer tires transmit more road input to the suspension and cabin. Reducing tire stiffness slightly can help in better filtering high-frequency road irregularities but handling and maneuverability should also be taken into consideration while doing this.

Scalability in a real industrial setting

A low-cost data logger fixed to the seat, right next to the driver's arm, has grabbed clear, repeatable vibration readings on any truck and any road. Because the hardware is simple and cheap, it slips smoothly into the building process at the factory. We judge the comfort level with ISO 2631, so the numbers carry real weight. Thanks to the modest sensor price, the easy mounting, and the direct value those numbers give engineers who tune seat suspensions, the whole setup is ready for large-scale roll-out. It pays off right away—drivers feel better—and it feeds long-term product development. The engineering job we tackle — measuring and boosting driver comfort through seat-vibration checks based on ISO 2631 — matters a lot to OEMs such as Volvo Trucks, Scania, MAN, and Daimler Trucks for several clear reasons:

- Ride comfort sells trucks: Fleet buyers know that a smoother ride cuts fatigue, lifts safety, and keeps drivers happy on long hauls, so comfort is a big selling point.
- Real-world data gap: OEMs already tweak seat and cab suspensions, yet they rarely own long-term, in-service vibration data from the full mix of roads and climates their trucks see.
- ISO 2631 links comfort to compliance: Using a worldwide standard means the work lines up with regulations and marketing claims everywhere.

This system scales especially well for OEMs because:

- Predictable data quality: Each logger bolts to the same spot—usually the seat base—so every truck feeds back clean, comparable data.
- Passive data capture: The truck keeps running its normal routes; nothing stops or changes, so fleets can gather millions of kilometers of real-world vibration without fuss.
- More miles, more insight: When thousands of trucks share vibration traces from every road class, climate zone, cargo load, and driving style, OEMs gain an unmatched picture of seat performance over time and space.

OEM engineering teams can turn that huge data pool into future gains by:

- Fine-tuning suspensions in the real world: Designs improve using actual vibration exposure, not just lab rigs.
- Forecasting wear and tear: Over the long haul, engineers can predict how comfort fades as foams compress and springs tire out.
- Targeted regional tuning: Seats and cabs can be optimized for the duty cycle—say, cross-continent highway versus stop-and-go city delivery.
- Building smart seats: Next-generation seats with adaptive or active damping can be taught to react to the true vibration patterns trucks face every day.

Conclusion

We had the privilege to co-ride a Volvo FH16 for about 30 km and placed a small 3-DOF logger (an accelerometer in a box) at the driver's seat armrest to catch every vibration on smooth, mixed and rough roads to understand the driver comfort for a long drive.

- On the smooth highway, the model was about 5% off (slightly low).
- On the mixed road, it was again about 5% off.
- On the rough track, the model read about 35% too high.

Even on the rough stretch, the real seat peaks in the critical 4 – 8 Hz band stayed below the ISO-2631 comfort limit of 0.33 g, so the driver remained in the comfort zone.

The model already works and is easy to scale for future seat designs, but it needs a bit more tweaking to hit the mark on rough roads and unlock even better long-haul comfort.

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 - Damping characteristics of seat cushion materials for tractor ride comfort
https://www.academia.edu/50035696/Damping_characteristics_of_seat_cushion_materials_for_tractor_ride_comfort

Appendix

Quarter-Car Parameters (Heavy Duty Truck) used in calculations:

Parameter	Value (approximate)
Driver seat mass (M_{ds})	100 kg * 
Sprung mass (M_s and M_{se})	3,600 kg * 
Unsprung mass (M_u and M_{ue})	300 kg * 
Suspension damping (C_s)	12,000 N·s/m * 
Seat damping (C_{ds})	800 N·s/m * 
Tyre stiffness (K_t)	1,800,000 N/m * 
Suspension stiffness (K_s)	410,000 N/m * 
Seat spring stiffness (K_{ds})	20,000 N/m * 